Method	Thomas $\mathcal{L}$	% err	Nested-Thomas $\mathcal{L}$	% err
ParamNet‡	0.125 ± 0.007	26%	0.192 ± 0.010	27%
vihrs ($L(r) - r$ CNN)	0.115 ± 0.007	25%	0.217 ± 0.006	29%
Minimum contrast, K	0.855 ± 0.682^{a}	83%	—^{d}	—
Minimum contrast, g	6.199 ± 0.339^{b}	407%	—^{d}	—
Chance-level reference (label shuffle)	1.003 ± 0.019	*92%*	1.006 ± 0.020	*72%*

Table 1: Estimator comparison on Thomas and Nested Thomas, $n = 10$ seeds, on the shared test split. % err is the approximate relative error in original parameter space, $100(\exp(\sqrt{\mathcal{L}}\,\bar{\sigma}_{\log}) - 1)$, with $\bar{\sigma}_{\log}$ the across-parameter average of the log-std used to normalize each process's labels ($\bar{\sigma}_{\log}^{\text{Thomas}} = 0.652$, $\bar{\sigma}_{\log}^{\text{Nested}} = 0.541$); since $\mathcal{L}$ aggregates parameters of unequal difficulty this is a rough single-number summary, see Table 2 for the per-parameter version. ‡ParamNet is process-specific: DTM_5 for Thomas, fused $\text{DTM}_{5,10,15}$ for nested Thomas. ^{a} Minimum contrast on K is unstable across seeds rather than uniformly bad, and ^{b} minimum contrast on g is consistently worse than chance. ^{d}Not run: no closed-form summary statistic for nested Thomas has yet been derived.

(a) Thomas (κ parent intensity, μ offspring intensity, σ cluster scale)

Parameter	ParamNet	% err	vihrs	% err	Min. contrast, K	% err	Min. contrast, g	% err
$\mathcal{L}^{\kappa}$	0.201 ± 0.010	30%	0.192 ± 0.013	29%	1.373 ± 1.087	97%	8.825 ± 0.485	455%
$\mathcal{L}^{\mu}$	0.134 ± 0.007	29%	0.122 ± 0.007	27%	0.939 ± 0.758	95%	6.334 ± 0.370	468%
$\mathcal{L}^{\sigma}$	0.040 ± 0.004	15%	0.029 ± 0.002	12%	0.254 ± 0.203	41%	3.437 ± 0.170	258%

(b) Nested Thomas

Parameter	ParamNet^{c}	% err	vihrs	% err
parent intensity	0.283 ± 0.023	30%	0.320 ± 0.010	32%
meta-offspring	0.402 ± 0.022	30%	0.493 ± 0.022	33%
meta-cluster scale	0.143 ± 0.006	27%	0.132 ± 0.009	26%
mean offspring	0.086 ± 0.004	16%	0.110 ± 0.010	19%
cluster scale	0.044 ± 0.000	15%	0.028 ± 0.003	12%

Table 2: Per-parameter loss $\mathcal{L}^{j}$ underlying the aggregate $\mathcal{L}$ of Table 1 (same methods, same seeds; ParamNet/nested is n=3). % err is $100(\exp(\sqrt{\mathcal{L}^{j}}\,\sigma_{\log}^{j})-1)$ with the per-parameter $\sigma_{\log}^{j}$ read off the saved (train-split) label normalization stats: Thomas $(\kappa, \mu, \sigma) = (0.577, 0.690, 0.688)$; Nested Thomas (parent, meta-offspring, meta-cluster scale, mean offspring, cluster scale) $= (0.492, 0.410, 0.627, 0.518, 0.656)$. Minimum contrast has no nested-Thomas columns.

Axis	Arm	Thomas $\mathcal{L}$	% err	Nested-Thomas $\mathcal{L}$	% err
Filtration	Rips	—[a]	—	—[a]	—
	Rips (vec_multik, landscape)	0.178 ± 0.008	32%	0.461 ± 0.016	44%
	DTM_5 / DTM_{15}	0.125 ± 0.007 / 0.146 ± 0.006	26% / 28%	0.205 ± 0.003 / 0.260 ± 0.017	28% / 32%
	$\mathrm{DTM}_{\{5,10,15\}}$ (fused)	0.126 ± 0.008	26%	0.192 ± 0.010^c	27%
Homology	H_0 only / H_1 only	0.125 ± 0.006 / 0.900 ± 0.019	26% / 86%	$0.280 \pm 0.250^\dagger$ / 0.932 ± 0.028	33% / 69%
Encoder	shared / independent, concat	0.125 ± 0.007 / 0.134 ± 0.009	26% / 27%	0.201 ± 0.010 / 0.222 ± 0.021	27% / 29%
Vectorization	Persistence image	0.126 ± 0.008	26%	0.192 ± 0.010^c	27%
	Landscape	0.132 ± 0.008	27%	0.250 ± 0.005	31%
	Silhouette ($p = 1$)	0.165 ± 0.013	30%	0.271 ± 0.012	33%
	Betti curve	0.196 ± 0.014	33%	0.290 ± 0.005	34%
Features	$\log N$ removed	$0.217 \pm 0.278^\dagger$	36%	0.203 ± 0.007	28%
	Topology removed ($\log N$ only)	0.893 ± 0.015	85%	0.948 ± 0.019	69%
Calibration	Naive ($q = 1.00$)	0.122 ± 0.003	26%	0.214 ± 0.012	28%

Table 3: Branch summary: every axis measured against the multi-scale, H_0+H_1, CoordConv starting configuration, at the default $q = 0.95$ calibration, $n = 10$ seeds per arm unless noted. % err uses the same aggregate, average-$\sigma_{\log}$ approximation as Table 1 ($\bar{\sigma}_{\log}^{\mathrm{Thomas}} = 0.652$, $\bar{\sigma}_{\log}^{\mathrm{Nested}} = 0.541$); rows marked † have at least one non-converged seed collapsed to chance level, so both the $\mathcal{L}$ and the % err for that arm are dominated by the failed seed(s) rather than being representative. [a] Rips fails on 10/10 seeds before training starts (native/CNN encoder path); the landscape row is the 1-D vec_multik path, which survives. [c] $n = 3$.